

Mathematical Model for Air Traffic Project

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Abstract

Brief explanation of the model implemented in the code.

1 Model Setup

I model directly the number of aeroplanes at an airport as a continuous time Markov chain (CTMC). A quick glance at the plots of this gathered data show that there are strong seasonal effects, predominantly along a 24h cycle, and so some kind of time inhomogeneous model with periodic intensities seems appropriate. In general, denoting number of planes at time t by X_t , the model can be expressed the local expression

$$\mathbb{P}(X_{t+\epsilon} = j | X_t = i) = \delta_{ij} + q_{ij}(t)\epsilon + o(\epsilon). \quad (1)$$

I opt to use a fixed functional form for the *intensities* $q_{ij}(t)$, namely piecewise constant and 24h periodic. Hence let $\xi = (\xi_0, \dots, \xi_{K-1})$ where $0 \leq \xi_0 < \xi_1 < \dots < \xi_{K-1} < 1440$, where note that 1440 is the number of minutes in a day (all the data I gather has this precision). Then for each i, j , let

$$q_{ij}(t) = \sum_{k=0}^{K-1} a_{ijk} \mathbb{1}_{\Xi_k}(t) \quad (2)$$

where $\Xi_k = \{t : t \bmod 1440 \in [\xi_k, \xi_{k+1})\}$ for $k = 0, \dots, K-2$ and $\Xi_{K-1} = \{t : t \bmod 1440 \in [\xi_{K-1}, \xi_0 + 1440)\}$.

Two advantages of this choice are that on each interval the model is a classic time homogenous CTMC and so the parameters can be easily interpreted and computations easily made, and second that given the ξ_k it is easy to solve for the maximum likelihood values for the coefficients a_{ijk} . A downside is that the model and corresponding likelihood function is not continuous in the ξ_k , and so optimising for the ξ_k is harder.

2 Training the Model

A natural approach to finding the model parameters is via maximum likelihood. Let $0 = T_0 < T_1 < \dots < T_N$ be the observed jump times and $Y_0, Y_1, \dots, Y_N$ be the corresponding states, i.e. number of planes at the airport. We compute the log likelihood

by

$$\begin{aligned}
\ell(a_{ijk}, \xi_k; Y_n, T_n) &= \sum_{n=1}^N \log(q_{Y_{n-1}Y_n}(T_n)) + \int_{T_{n-1}}^{T_n} q_{Y_{n-1}Y_n}(t) dt \\
&= \sum_{n=1}^N \log(q_{Y_{n-1}Y_n}(T_n)) - \int_{T_{n-1}}^{T_n} \sum_{z \neq Y_{n-1}} q_{Y_{n-1}z}(t) dt \\
&= \sum_x \sum_{y \neq x} \sum_{i \in S_{xy}} \log(q_{xy}(T_i)) - \int_{T_{i-1}}^{T_i} \sum_{z \neq x} q_{xz}(t) dt
\end{aligned}$$

where $S_{xy} = \{i : Y_{i-1} = x, Y_i = y\}$. Filling in the chosen functional form for q gives

$$\begin{aligned}
\ell(a_{ijk}, \xi_k; Y_n, T_n) &= \sum_x \sum_{y \neq x} \sum_{i \in S_{xy}} \sum_{k=0}^{K-1} \log(a_{xyk}) \mathbb{1}_{\Xi_k}(T_i) - \sum_{z \neq x} a_{xzk} \int_{T_{i-1}}^{T_i} \mathbb{1}_{\Xi_k}(t) dt \\
&= \sum_x \sum_{y \neq x} \sum_{k=0}^{K-1} \log(a_{xyk}) \chi_{xyk} - a_{xyk} \Delta_{xk}
\end{aligned}$$

where $\chi_{x,y,k} = \sum_{i \in S_{xy}} \mathbb{1}_{\Xi_k}(T_i)$, which is the number of observed jumps from x to y during the time windows of Ξ_k , and $\Delta_{xk} = \sum_{z \neq x} \sum_{i \in S_{xz}} \int_{T_{i-1}}^{T_i} \mathbb{1}_{\Xi_k}(t) dt$, which is the total time spent in state x during the time windows of Ξ_k .

2.1 Optimising a

For fixed ξ_k , it is then easy to compute the maximum likelihood estimates for a_{ijk} as

$$\hat{a}_{ijk} = \frac{\chi_{ijk}}{\Delta_{ik}}. \quad (3)$$

Note it is exactly the choice of functional form that allows the sum inside the logarithm to move outside which allows this to be done in closed form.

2.2 Optimising ξ

ℓ depends discontinuously on the ξ_k and so some kind of brute force search is needed. Currently implemented in the code is a kind of 'simulated annealing' style optimisation, picking one of the ξ_k at random then testing a random sample of possible values for it and choosing the one with the maximum likelihood. The sampling process is designed to encourage early on the ξ_k to move a lot to find a good value, and then to slow down its movement.

I plan to investigate alternative optimisation algorithms. A natural one to try is to view the ξ_k as model hyperparameters and use cross-validation to select them.

2.3 Restricted Parameter Models

An extension I have recently added allows for some extra structural restrictions to be added to the parameters of the model. In particular, it allows for some a_{ijk} to be set either to 0, or to be set equal to each other. This can be used for example to test if a queueing type model is a reasonable approximation. The above computations hold with very minor adjustments, leading to the maximum likelihood estimate using a pooled version of formula 3.

3 Measuring Model Error

Maximum likelihood is a convenient and theoretically sensible way to fit parameters, but it is good to also have a more natural measure of error to measure the model with. A natural error measure would seem to be

$$E(a, \xi; x) := \mathbb{E} \left(\frac{1}{T} \int_0^T |X_t - x_t| dt \right) \quad (4)$$

where x_t is the observed process and the expectation is with respect to the random process X_t . This is not easy to access directly, but can be estimated by obtaining N samples $X_t^1, \dots, X_t^N$, to give

$$\hat{E}(a, \xi; x) := \frac{1}{N} \sum_{n=1}^N \frac{1}{T} \int_0^T |X_t^n - x_t| dt. \quad (5)$$

4 Transition Semi-group

Once the model is trained, it is possible to then solve the *Kolmogorov Forward Equations* to obtain the transition semi-group $P(t) = (P_0(t), \dots, P_{S-1}(t))$ for given initial probability distribution, where S is the total number of observed states. This is the system of ODEs

$$P'(t) = P(t)Q(t)$$

The code currently implements this using an implicit Euler scheme solved separately over each interval where Q is constant. The scheme is

$$(1 - hQ)^T P_{n+1} = P_n$$

where h is the step size and is solved using standard linear algebra techniques, namely using QR decomposition as we repeatedly solve a linear system with the same coefficient matrix. First note that this scheme preserves the condition that the probabilities sum to 1 by multiplying on the left by the constant vector of ones. Moreover this scheme is stable as you can show that the magnitude of the eigenvalues of $(1 - hQ)^T$ are greater than or equal to one.

Access to this semi-group allows for quantitative predictions to be made and to test for deviation from the model.